Temporal Graph Models for Bankruptcy Prediction: A Fiscal-Year Evaluation

Atharv Shanker, Vedang Pandey, and Shoumik Mitra

Abstract. We propose Delta, a GCN–LSTM model that combines a firm's recent financial history with financial-peer information. We compare Delta with tabular, recurrent, graph, and score-fusion baselines on a public SEC-derived dataset using fiscal-year splits. Delta achieves average precision (AP) 0.079 and ROC-AUC 0.771. XGBoost remains the strongest individual model, with AP 0.114 and ROC-AUC 0.887. Exact filing dates are unavailable, so results are interpreted by fiscal year.

I. Introduction

Corporate bankruptcy is costly to firms, investors, employees, and lenders. Predictive models must find a small number of future events among many non-events, making average precision (AP) and recall at an operating point more informative than accuracy. Prior work on publicly traded U.S. firms reports that gradient-boosted trees are strong one-year-ahead predictors, establishing a demanding tabular baseline [1]. Graph neural networks (GNNs) have also been applied to bankruptcy and distress prediction, including models that combine corporate relations with temporal encoders [2], [3].

This study evaluates whether financial-peer graph messages improve bankruptcy ranking beyond firm-level temporal and boosted-tree models. Delta encodes up to five years of company features with an LSTM, then applies graph convolution over a financial-similarity peer graph. The evaluation includes tuned XGBoost and LightGBM baselines, conventional graph models, a reliability-aware temporal GNN, and validation-selected score fusion. Published results are reported separately when their datasets and evaluation protocols differ.

II. Related Work

Altman introduced a discriminant score for bankruptcy prediction [4]. Alanis et al. benchmarked eight methods on U.S. public firms using walk-forward one-year-ahead forecasts and report ROC-AUC 0.92 for XGBoost [1]. Table II lists this result separately because the cohort, predictors, and protocol differ from those used here. Zhao et al. combine intra-risk and contagion-risk signals with graph learning [2]. Sun et al. combine relation-specific graph convolution, relation attention, temporal encoding, and score fusion for distress prediction [3]. These studies establish temporal and relational graph learning as prior approaches to financial distress prediction.

The American Stock Market bankruptcy release initially considered for this project has a target constant within every company across its observed years, so it cannot support the intended next-year forecasting task as released [5]. We instead use the public SEC-derived numeric extension and its inherited next-year bankruptcy labels [6]. The thesis that introduced this extension reports stronger AP for its FinBERT text model than for its numeric-only variants; those published figures are context, not results from our run, and are not directly comparable with our numeric-only experiment.

III. Data and Task

A. Source and Cohort

The source contains filing-level rows joined by company identifier and SEC accession, a binary label, a year, and 34 dense numeric financial concepts. Its reported full cohort is 42,956 filings from 2008–2021. We audited the downloaded dense variant and found 214 rows belonging to repeated company-year groups (103 groups); one repeated group had conflicting labels. Since the release does not provide filing dates, selecting an amendment or duplicate as the canonical filing would require an unsupported temporal assumption. We therefore exclude every row in a repeated company-year group. The resulting cohort contains 42,742 observations, 5,612 firms, 293 positive observations (0.6855%), and 34 numeric predictors. Source rows and license files are not redistributed by this project; readers should download them from the repository record and cite the dataset author [6].

The source defines its inherited target as bankruptcy in the next fiscal year. Exact filing or publication dates are unavailable, and 103 repeated company-year groups cannot be ordered by fiscal year alone. The evaluation therefore uses fiscal-year splits and does not verify whether each feature was available on a specific prediction date.

B. Chronological Split

We train on fiscal years 2008–2016, select models and fusion weights on 2017–2018, and evaluate once on 2019–2021. The test set contains 9,032 rows and 64 positive labels (0.7086% prevalence). Company identifiers occur in multiple years, so the primary split tests future years for a partially overlapping issuer population; the present report does not claim company-disjoint generalization. No test labels are used for feature transforms, model selection, or threshold selection.

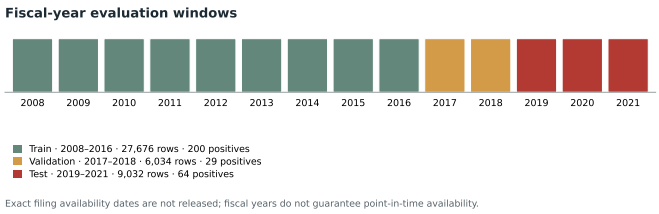


Fig. 1. Fiscal-year evaluation windows. The source does not provide exact filing-availability dates, so the split is chronological by fiscal year rather than verified filing time.

IV. Methods

A. Preprocessing and Strong Tabular Baselines

Numeric features are converted to finite values, winsorized at the training-set 1st and 99th percentiles, median-imputed, and scaled by the training interquartile range. All transformation parameters are fitted on training years. We compare logistic regression, decision tree, random forest, histogram gradient boosting, MLP, LSTM, GCN, GraphSAGE, reliability-aware temporal GNN, LightGBM, and XGBoost. For each GBDT family we evaluate eight declared configurations on both the most recent-year input and flattened five-year history (16 validation trials), then select by validation AP. LightGBM selects 300 trees, learning rate 0.03, seven leaves, maximum depth four, 20 minimum child samples, and ℓ_2 penalty 1.0, with balanced class weighting. XGBoost selects depth two, learning rate 0.03, 300 trees, minimum child weight 10, ℓ_2 penalty 1.0, row and feature subsampling 0.8, and training-set positive-class weighting.

B. Delta: GCN with an LSTM Encoder

Delta encodes a trailing sequence of up to five annual feature vectors for firm i through an LSTM, producing state $\mathbf{h}_{i,t}$. At prediction year t, we construct a k-nearest-neighbor financial-similarity graph ($k = 10$) from the standardized financial features. The graph convolution combines each firm's temporal state with the similarity-weighted mean of its neighbors' states, followed by a binary classifier for next-year bankruptcy. Delta uses the conventional peer graph without relation-persistence weighting. Financial-similarity links represent predictive associations, not causal or contagion relationships.

C. Reliability-Aware Temporal GNN Comparator

For firm i in year t, an LSTM maps a trailing sequence of up to five annual feature vectors to state $\mathbf{h}_{i,t}$. For each financial feature group r, we construct a directed k-nearest-neighbor graph ($k = 10$) using cosine similarity of the corresponding standardized features in year t. For edge (j, i), the base weight is $s_{ji,t} = \max(0.01, 1 - d_{\cos}(\mathbf{x}_{j,t}, \mathbf{x}_{i,t}))$. We down-weight unstable neighborhoods using the Jaccard overlap $p_{i,t}$ between firm i's peer sets in adjacent snapshots:

$$w^{(r)}_{ji,t} = s^{(r)}_{ji,t}\left(0.5 + 0.5p^{(r)}_{i,t}\right). \tag{1}$$

Each relation channel computes a weighted mean of neighboring temporal states. A learned softmax gate combines relation messages, which are fused with the firm's own state and passed to a binary classifier. The GCN and GraphSAGE comparators use the same LSTM states and contemporaneous peer graph, with a single merged message channel. Financial similarity edges are predictive associations; they do not establish contagion or causal links.

D. Validation-Selected Score Fusion

We transform the tree and graph scores with robust location/scale parameters fitted on validation data, then evaluate convex combinations $q = (1 - \alpha)q_{\text{tree}} + \alpha q_{\text{graph}}$ for $\alpha \in \{0, 0.1, \ldots, 1\}$. We choose the combination with highest validation AP subject to validation ROC-AUC not falling below the tree score. The same validation-only rule selects $\alpha = 0.2$ for LightGBM–GraphSAGE, $\alpha = 0.1$ for LightGBM–reliability-GNN, and $\alpha = 0.1$ for XGBoost–reliability-GNN. XGBoost–GCN and XGBoost–GraphSAGE select $\alpha = 0$, so validation does not support adding either message-passing score under this rule.

E. Metrics and Uncertainty

AP is the primary metric and ROC-AUC is co-required. We also report Brier score for individual probability models and precision/recall at a threshold chosen on validation to target 80% recall. Fused standardized scores are rankings, not calibrated probabilities; we do not report Brier scores for them. Paired 95% confidence intervals compare test predictions by resampling companies with replacement and retaining all test-year rows for each sampled company (2,000 valid replicates). One training seed (42) and one future test window are available, so this interval does not capture seed or period variability.

V. Experimental Results

Table II preserves the published XGBoost benchmark value. It is context only, not a matched test-set result.

XGBoost is the strongest individual baseline in this evaluation on both AP and ROC-AUC, with AP 0.1141 and ROC-AUC 0.8868. Delta achieves AP 0.0794 and ROC-AUC 0.7708. Its AP is 0.0086 higher than the LSTM's, while its ROC-AUC is 0.0032 lower. The reliability-aware temporal GNN comparator achieves AP 0.0383 and ROC-AUC 0.7515. Each neural model uses one training seed. Figure 3 visualizes every reported table metric, while Figure 4 shows ranking curves for the tabular baselines and temporal graph models.

The validation-selected LightGBM–GraphSAGE blend increases test AP from 0.1116 to 0.1181, while ROC-AUC decreases from 0.8790 to 0.8705. The paired company-clustered AP difference is 0.00515 with 95% interval [–0.00285, 0.01913]; the ROC-AUC difference is –0.00869 [–0.01783, –0.00065]. For XGBoost, validation selects a GNN weight of 0.1. The fused test AP is 0.1133 and

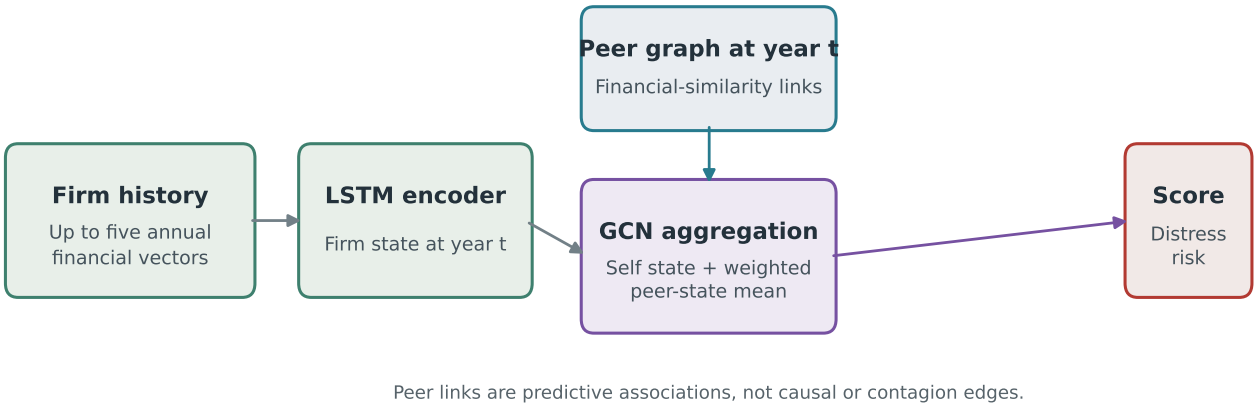


Fig. 2. Delta combines an LSTM encoding of up to five annual financial vectors with graph convolution over the financial-similarity peer graph at year t. The GCN combines a firm's temporal state with a similarity-weighted mean of peer states before predicting distress risk.

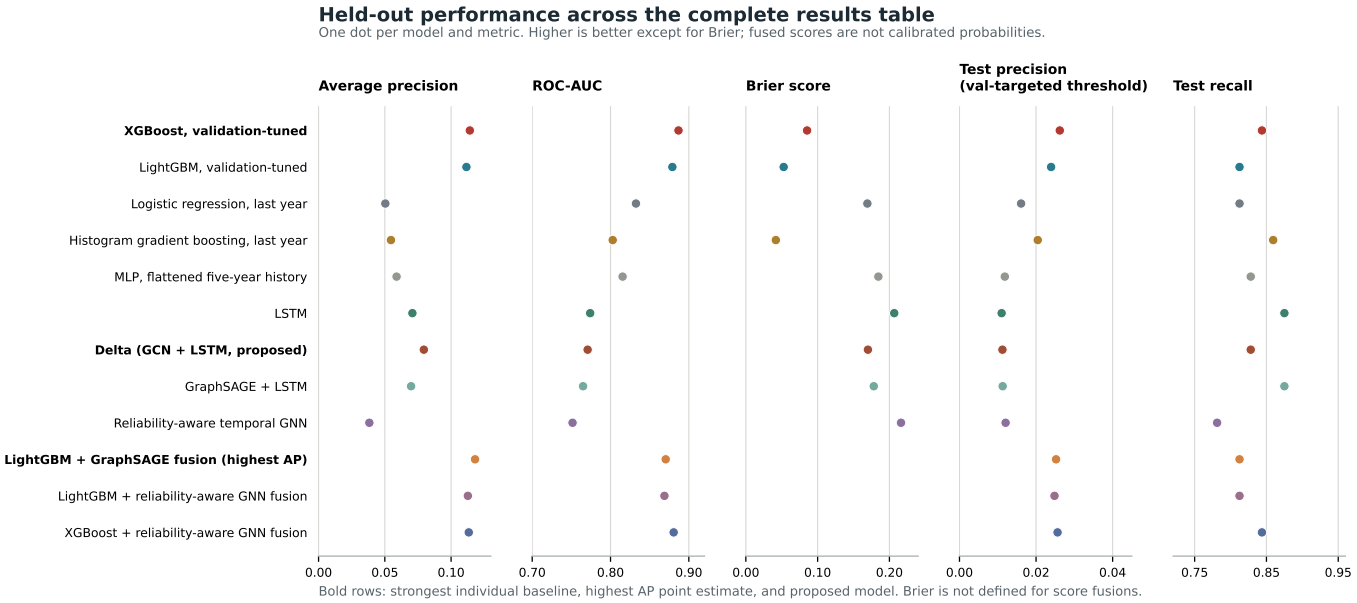


Fig. 3. All metrics in Table I, shown as metric-specific panels on their native scales. Higher is better except for Brier score. Score-fusion rows have no Brier score because the fused ranking scores are not calibrated probabilities. The strongest individual baseline and proposed model are bolded.

ROC-AUC is 0.8806, compared with 0.1141 and 0.8868 for XGBoost alone. The paired AP-difference interval is [–0.01363, 0.01493], and the ROC-AUC difference is –0.00626 [–0.01558, –0.00037].

An additional diagnostic adds 12 conventional financial ratios to the 34 source features. Ratio-augmented LightGBM scores AP 0.0827 and ROC-AUC 0.8670. The ratio-augmented reliability-GNN scores AP 0.0286.

The originating dataset thesis reports AP 0.08 and ROC-AUC 0.87 for its raw numeric XBRL representation, and AP 0.30 and ROC-AUC 0.82 for its text FinBERT system [6]. These results use the unreduced filing cohort and are included as source-reported context.

VI. Conclusion

Delta combines an LSTM representation of firm history with graph convolution over financial-similarity peers. Among individual models, XGBoost achieved the highest AP and ROC-AUC. Delta's AP exceeded that of the LSTM, while its ROC-AUC was slightly lower. The reliability-aware temporal GNN and its fusion with XGBoost scored below the tree model alone. The LightGBM–GraphSAGE blend had the highest AP point estimate but lower ROC-AUC than LightGBM; its paired AP

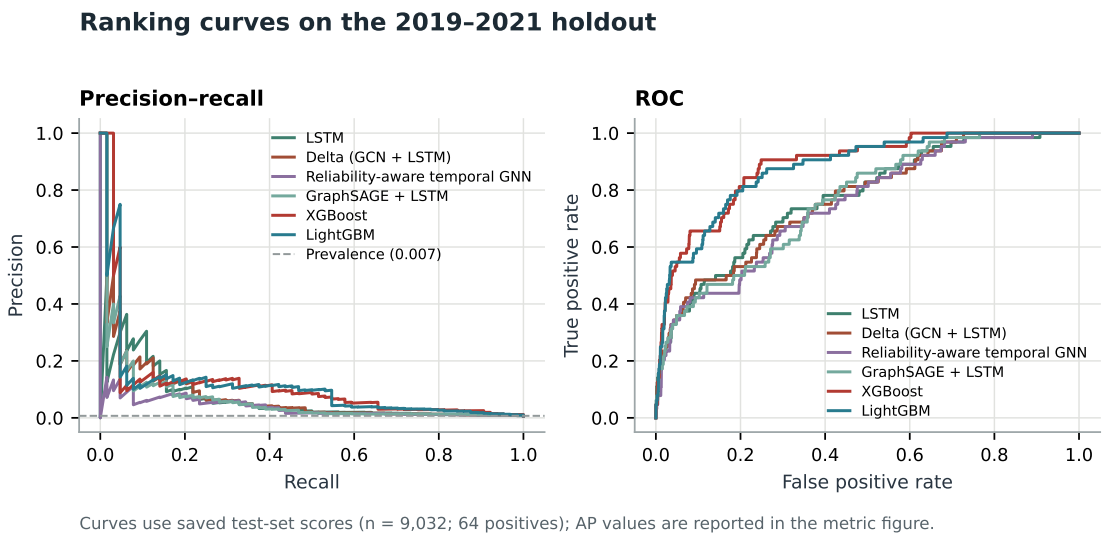


Fig. 4. Precision–recall and ROC curves for the strongest tabular baselines, LSTM, Delta (GCN–LSTM), GraphSAGE–LSTM, and the reliability-aware temporal GNN comparator on the held-out years. The dashed line in the precision–recall panel marks test prevalence. Curves use saved test-set predictions.

TABLE I
Held-out fiscal-year test results (2019–2021, $n = 9{,}032$, 64 positives). Thresholds target 80% recall on validation; precision and recall are measured on test. Bold marks the strongest individual baseline, highest AP point estimate, and proposed model.

Model	AP ↑	ROC-AUC ↑	Brier ↓	Test precision	Test recall
XGBoost (best individual baseline)	0.1141	0.8868	0.0853	0.0262	0.8438
LightGBM, validation-tuned	0.1116	0.8790	0.0527	0.0239	0.8125
Logistic regression, last year	0.0504	0.8325	0.1693	0.0161	0.8125
Histogram gradient boosting, last year	0.0546	0.8028	0.0418	0.0204	0.8594
MLP, flattened five-year history	0.0589	0.8154	0.1846	0.0118	0.8281
LSTM	0.0708	0.7740	0.2067	0.0110	0.8750
(Ours) Delta (GCN + LSTM)	0.0794	0.7708	0.1701	0.0112	0.8281
GraphSAGE + LSTM	0.0698	0.7649	0.1782	0.0113	0.8750
Reliability-aware temporal GNN	0.0383	0.7515	0.2162	0.0120	0.7812
LightGBM + GraphSAGE fusion (highest AP)	0.1181	0.8705	–	0.0252	0.8125
LightGBM + reliability-aware GNN fusion	0.1127	0.8688	–	0.0248	0.8125
XGBoost + reliability-aware GNN fusion	0.1133	0.8806	–	0.0256	0.8438

TABLE II
Published benchmark context from Alanis et al. The source reports ROC-AUC for XGBoost; AP is not reported. Different cohort, predictors, and walk-forward protocol make this non-comparable to our holdout.

Evaluation	AP	ROC-AUC
Alanis et al. [1]	n/r	0.9200
This study, XGBoost	0.1141	0.8868

confidence interval includes zero. Exact filing dates were unavailable, so these comparisons use fiscal-year ordering.

References

[1] E. Alanis, S. Chava, and A. Shah, "Benchmarking machine learning models to predict corporate bankruptcy," arXiv preprint arXiv:2212.12051, 2022. [Online]. Available: https://arxiv.org/abs/2212.12051

[2] Y. Zhao, S. Wei, Y. Guo, Q. Yang, X. Chen, Q. Li, F. Zhuang, J. Liu, and G. Kou, "Combining intra-risk and contagion risk for enterprise bankruptcy prediction using graph neural networks," Information Sciences, 2023.

[3] Z. Sun, F. Shao, and F. Meng, "Network-aware financial distress prediction via hierarchical attention on dynamic heterogeneous corporate networks," Procedia Computer Science, vol. 287, pp. 453–460, 2026.

[4] E. I. Altman, "Financial ratios, discriminant analysis and the prediction of corporate bankruptcy," The Journal of Finance, vol. 23, no. 4, pp. 589–609, 1968.

[5] G. Lombardo, M. Pellegrino, G. Adosoglou, S. Cagnoni, P. M. Pardalos, and A. Poggi, "Machine learning for bankruptcy prediction in the american stock market: Dataset and benchmarks," Future Internet, vol. 14, no. 8, p. 244, 2022.

[6] A. Potiriadis, "An open financial dataset for bankruptcy prediction," Master's thesis, University of Peloponnese, Jul. 2026. [Online]. Available: https://doi.org/10.26263/amitos-3539